

Hochiminh city University of Technology
Faculty of Computer Science and Engineering



COMPUTER GRAPHICS

CHAPTER 05:

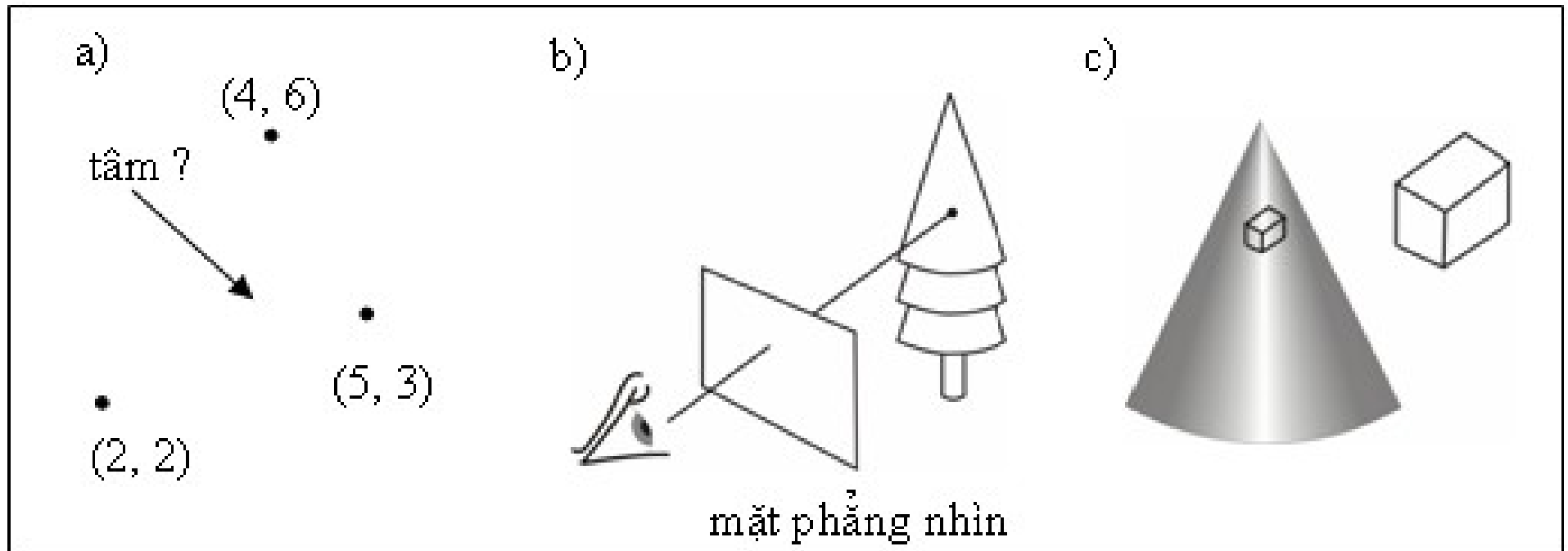
Vector in Computer Graphics

OUTLINE

- ☐ Vector
- ☐ Dot product
- ☐ Cross Product
- ☐ Scalars
- ☐ Points
- ☐ Affine Sums
- ☐ Parametric Form
- ☐ Line
- ☐ Plane
- ☐ Some Example
- ☐ Representation

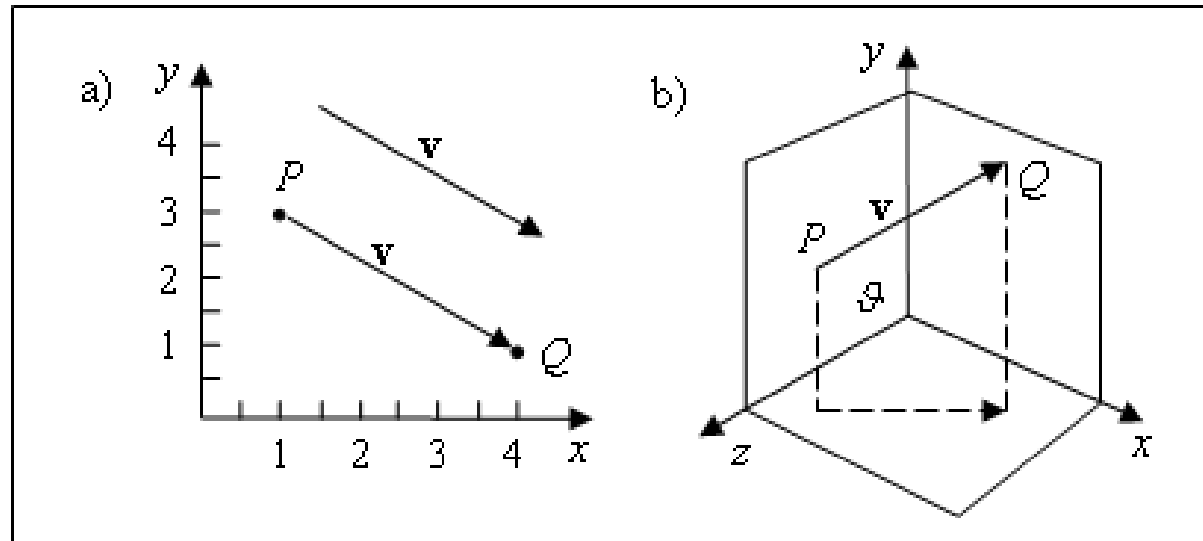
Vector

- ❑ Why vector important?
 - Remove Hidden-face.
 - Normal vector
 - Three basic elements in geometry: scalar, point, vector



Vector

- ❑ Physical definition: a vector is a quantity with two attributes: 1)**Direction**; 2)**Magnitude**
- ❑ Examples include: 1)**Force**; 2)**Velocity**
- ❑ Vectors Lack Position: These vectors are identical
 - **Same length and magnitude**



Vector

❑ $A = (1, 2, 9), B = (4, 6, 3)$

❑ $AB = (4 - 1, 6 - 2, 3 - 9) = (3, 4, -6)$

❑ $BA = (1 - 4, 2 - 6, 9 - 3) = (-3, -4, 6)$

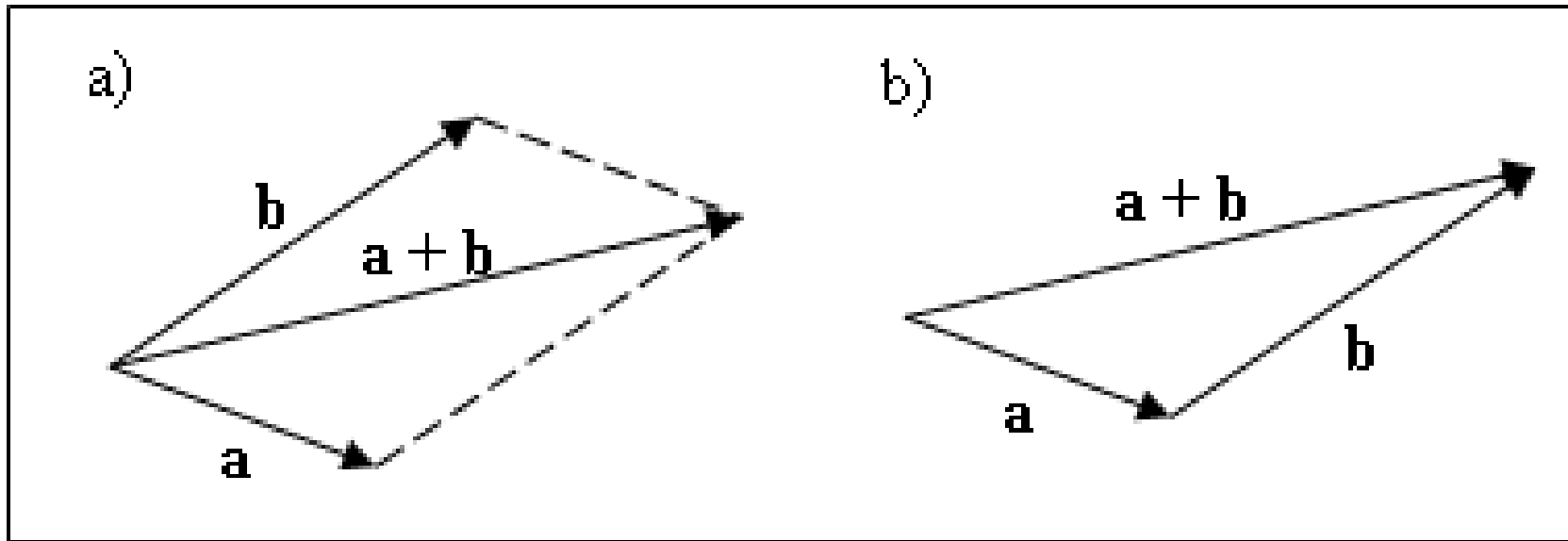
Vector

$$\mathbf{a} = (2, 5, 6), \mathbf{b} = (-2, 7, 1)$$

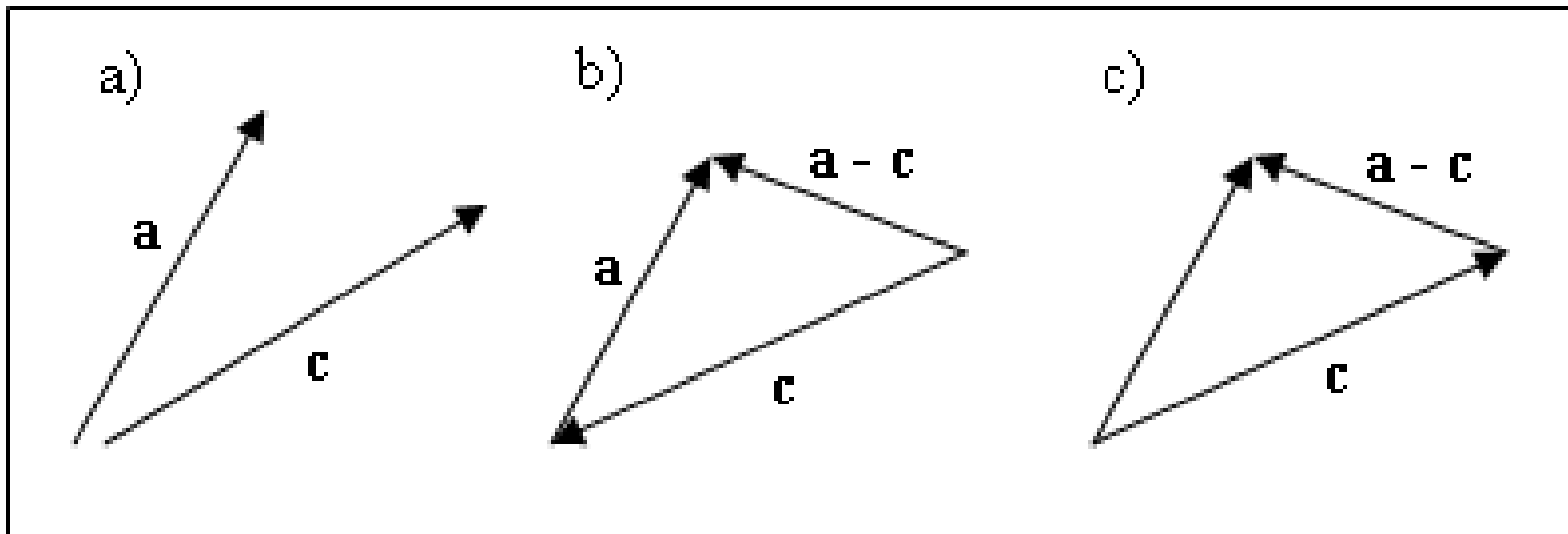
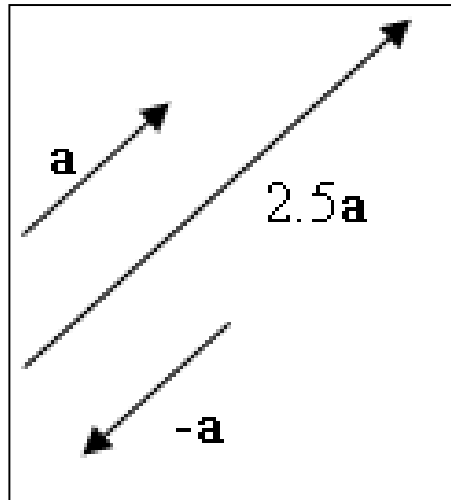
❑ Addition: $\mathbf{a} + \mathbf{b} = (0, 12, 7)$

❑ Scalar-vector multiplication: $6\mathbf{a} = (12, 30, 39)$

❑ Subtraction: $\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}) = (4, -2, 5)$



Vector



Vector

□ **Magnitude:** $|\mathbf{w}| = \sqrt{w_1^2 + w_2^2 + \dots + w_n^2}$

□ **Unit vector:** $\mathbf{u}_a = \frac{\mathbf{a}}{|\mathbf{a}|}$

Dot product

- Definition: The dot product d of two n -dimensional vectors $\mathbf{v} = (v_1, v_2, \dots, v_n)$ and $\mathbf{w} = (w_1, w_2, \dots, w_n)$ is denoted as $\mathbf{v} \bullet \mathbf{w}$ and has the value:

$$d = \mathbf{v} \bullet \mathbf{w} = \sum_{i=1}^n v_i w_i$$

□ Properties

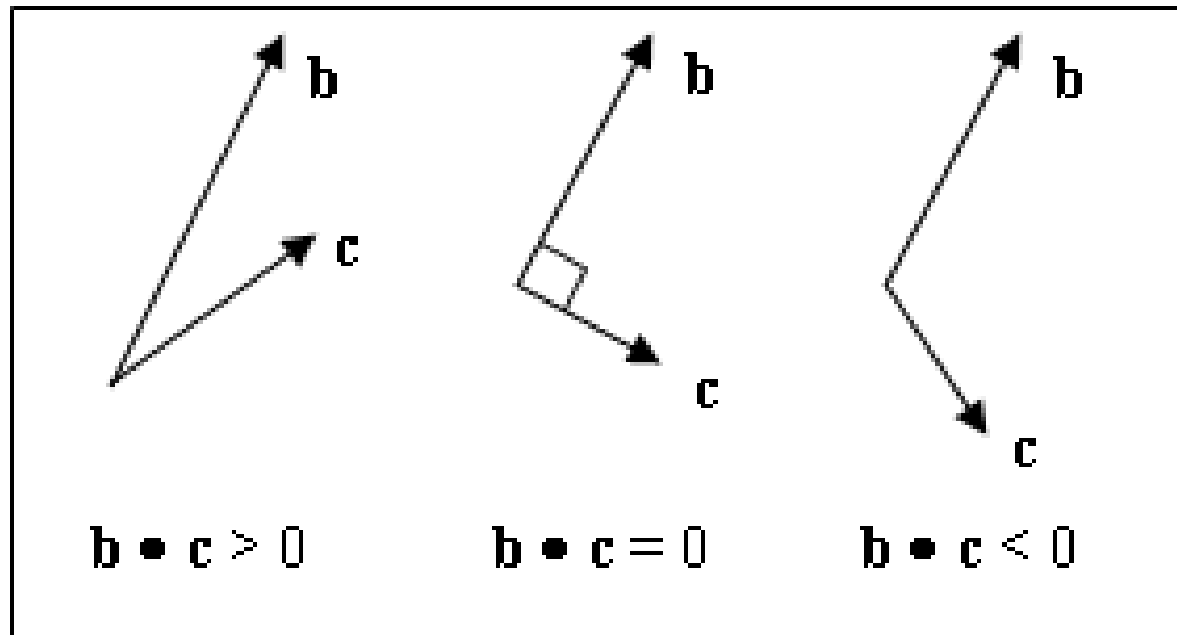
- Symmetry: $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$
- Linearity: $(\mathbf{a} + \mathbf{c}) \bullet \mathbf{b} = \mathbf{a} \bullet \mathbf{b} + \mathbf{c} \bullet \mathbf{b}$
- Homogeneity: $(s\mathbf{a}) \bullet \mathbf{b} = s(\mathbf{a} \bullet \mathbf{b})$
- $|\mathbf{b}|^2 = \mathbf{b} \bullet \mathbf{b}$

Dot product

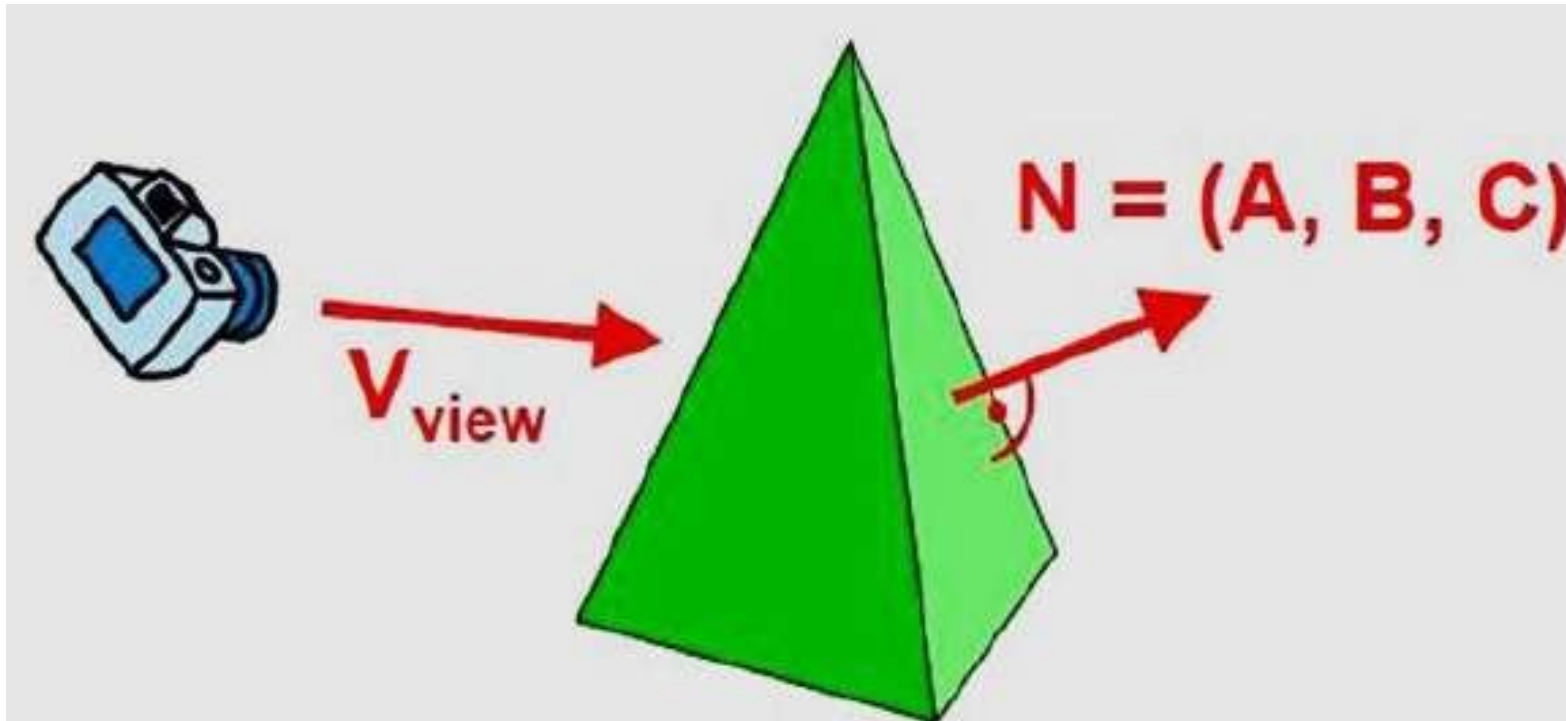
- The angle between two vectors

$$\mathbf{b} \bullet \mathbf{c} = |\mathbf{b}| |\mathbf{c}| \cos(\theta)$$

$$\cos(\theta) = \mathbf{u}_b \bullet \mathbf{u}_c$$



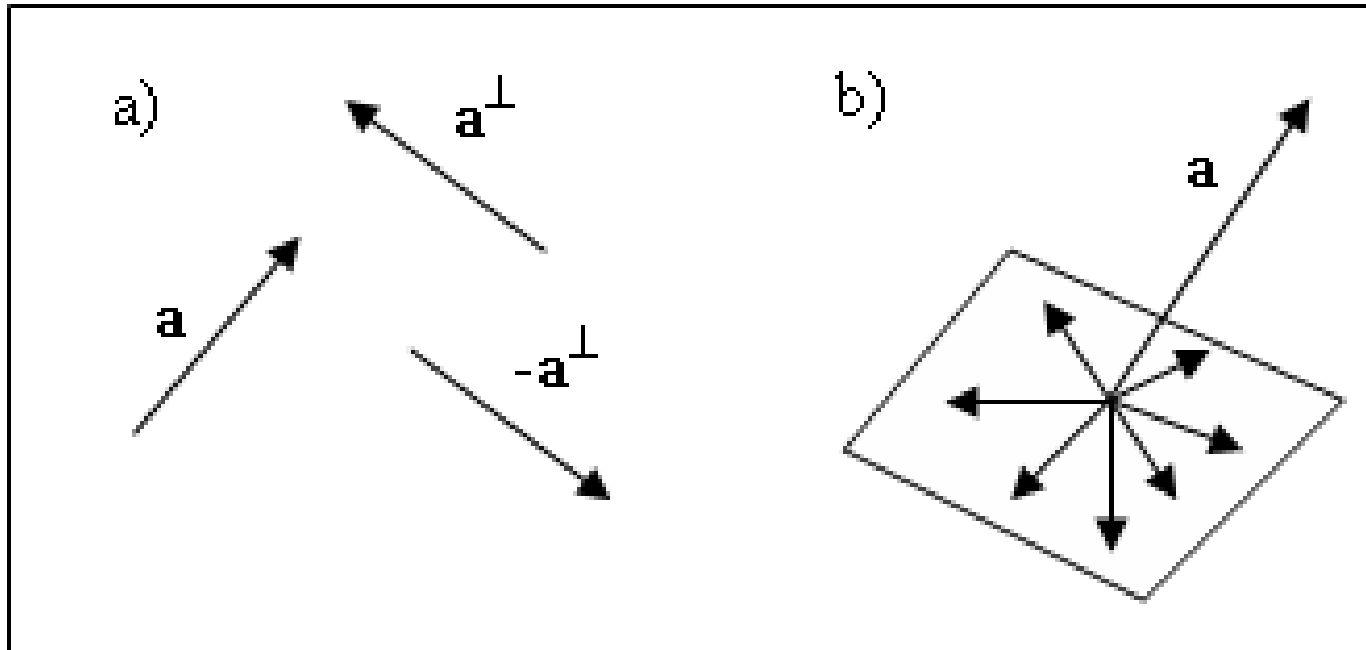
Dot product



Dot product

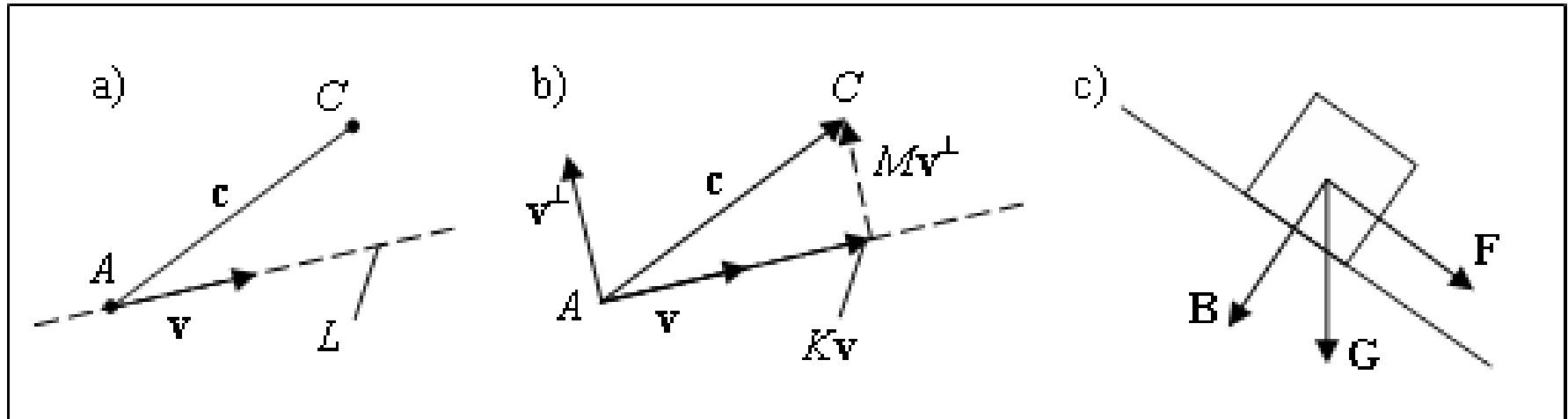
□ The 2D “perp” vector

- **Suppose** $\mathbf{a} = (a_x, a_y)$, then $\mathbf{a}^\perp = (-a_y, a_x)$ is the counterclockwise perpendicular to $\mathbf{a}$.



Dot product

□ Orthogonal Projections and the Distance from a point to a line



$$\mathbf{c} = K\mathbf{v} + M\mathbf{v}^\perp \quad (K, M = ?)$$

Dot product

$$\square \mathbf{c} = K\mathbf{v} + M\mathbf{v}^\perp \quad (K, M = ?)$$

$$(c_x, c_y) = K(v_x, v_y) + M(v_x^\perp, v_y^\perp)$$

$$\begin{cases} c_x = K^*v_x + M^*v_x^\perp \\ c_y = K^*v_y + M^*v_y^\perp \end{cases}$$

Dot Product

$$\square \mathbf{c} = K\mathbf{v} + M\mathbf{v}^\perp \quad (K, M = ?)$$

$$\mathbf{c} \bullet \mathbf{v} = K\mathbf{v} \bullet \mathbf{v} + M\mathbf{v}^\perp \bullet \mathbf{v} \rightarrow K = \frac{\mathbf{c} \bullet \mathbf{v}}{\mathbf{v} \bullet \mathbf{v}}$$

$$\mathbf{c} \bullet \mathbf{v}^\perp = K\mathbf{v}^\perp \bullet \mathbf{v} + M\mathbf{v}^\perp \bullet \mathbf{v}^\perp \rightarrow M = \frac{\mathbf{c} \bullet \mathbf{v}^\perp}{\mathbf{v}^\perp \bullet \mathbf{v}^\perp}$$

$$\mathbf{c} = \left(\frac{\mathbf{v} \bullet \mathbf{c}}{|\mathbf{v}|^2} \right) \mathbf{v} + \left(\frac{\mathbf{v}^\perp \bullet \mathbf{c}}{|\mathbf{v}|^2} \right) \mathbf{v}^\perp$$

$$\text{distance} = \left| \frac{\mathbf{v}^\perp \bullet \mathbf{c}}{|\mathbf{v}|^2} \mathbf{v}^\perp \right| = \frac{|\mathbf{v}^\perp \bullet \mathbf{c}|}{|\mathbf{v}|}$$

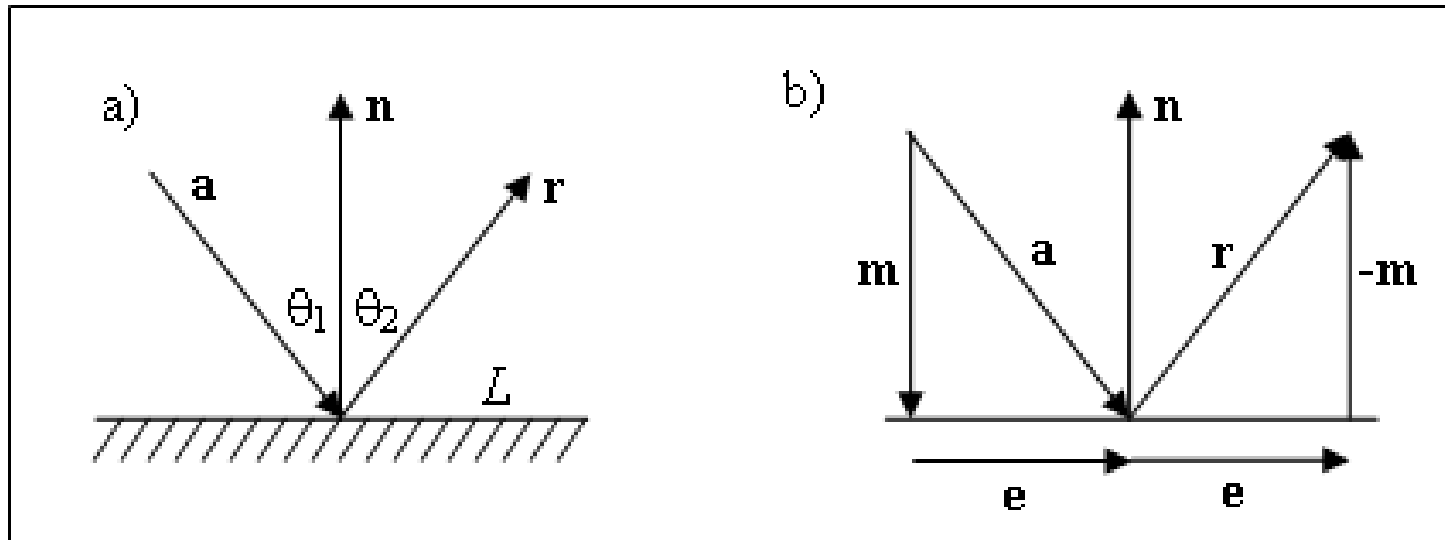
Dot product

□ Reflection

$$\mathbf{r} = \mathbf{e} - \mathbf{m}, \mathbf{e} = \mathbf{a} - \mathbf{m} \rightarrow \mathbf{r} = \mathbf{a} - 2\mathbf{m}$$

$$\mathbf{m} = \frac{\mathbf{a} \cdot \mathbf{n}}{|\mathbf{n}|^2} \mathbf{n} = (\mathbf{a} \cdot \mathbf{u}_n) \mathbf{u}_n$$

$$\mathbf{r} = \mathbf{a} - 2(\mathbf{a} \cdot \mathbf{u}_n) \mathbf{u}_n$$



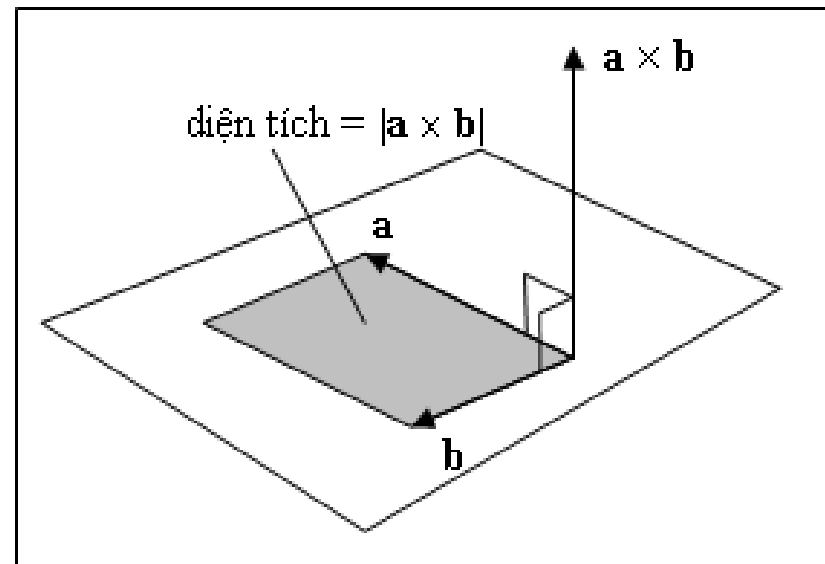
Cross Product

- ❑ The cross product of two vector is a vector
- ❑ Only for 3-dimensinal vector
- ❑ Suppose $\mathbf{a} = (a_x, a_y, a_z)$ and $\mathbf{b} = (b_x, b_y, b_z)$, then the cross product of a and b is

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y)\mathbf{i} + (a_z b_x - a_x b_z)\mathbf{j} + (a_x b_y - a_y b_x)\mathbf{k}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

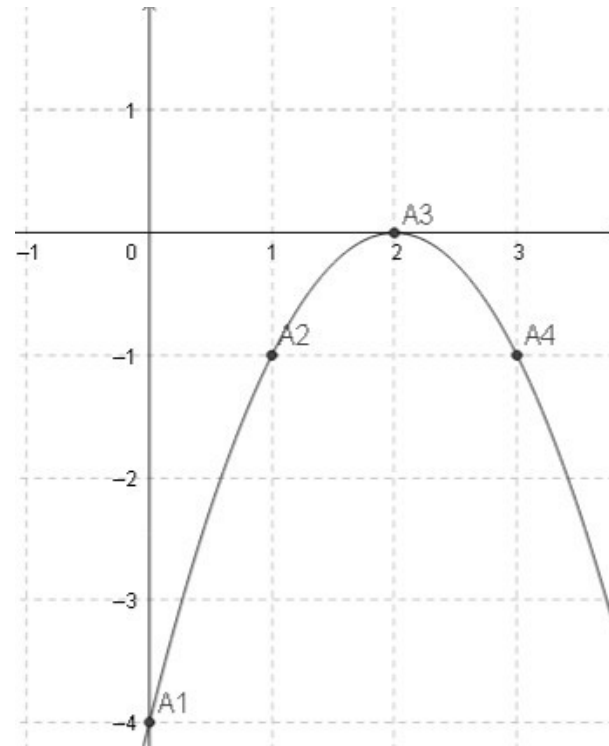
$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}||\mathbf{b}|\sin(\theta)$$



Parametric Form

□ Explicit

$$y = -x^2 + 4x - 4$$



□ How to draw

$$x^2 + y^2 = 1$$

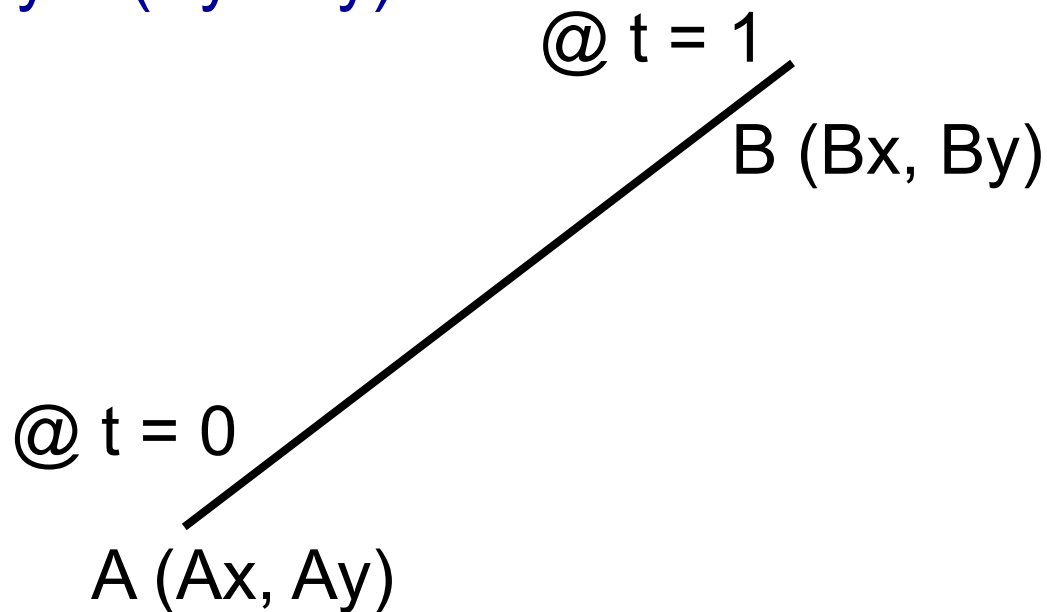
Parametric Form

□ Parametric form

- Ex 1: a straight line passes through A and B. Choose a parametric form that visit A at $t = 0$, visit B at $t = 1$.

$$x(t) = A_x + (B_x - A_x)t$$

$$y(t) = A_y + (B_y - A_y)t$$



Parametric Form

□ Parametric form

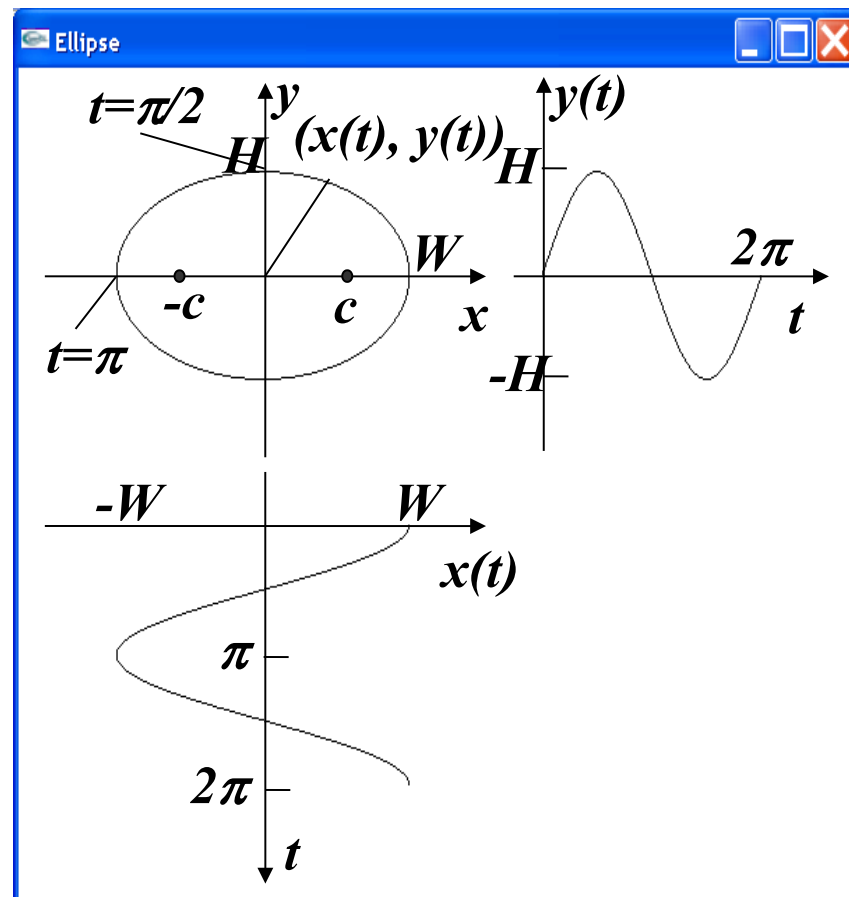
$$(x/W)^2 + (y/H)^2 = 1$$

EX 2: Ellipse with radius W and H

$$x(t) = W\cos(t)$$

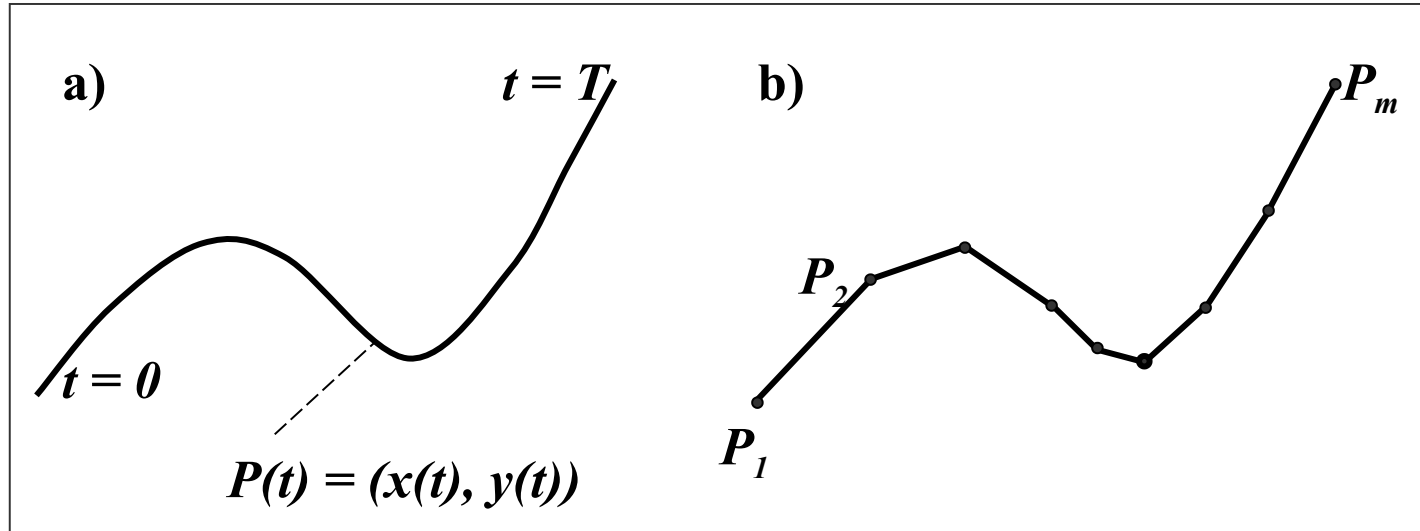
$$y(t) = H\sin(t)$$

với $(0 \leq t \leq 2\pi)$



Parametric Form

□ Draw parametric form curve



```
//draw the curve (x(t), t(t)) using  
//the array t[0], ..., t[n-1] of "sample-times"  
glBegin(GL_LINE_STRIP) ;  
    for(int i=0;i<n;i++)  
        glVertex2f(x(t[i]), y(t[i]));  
glEnd() ;
```

Parametric Form

❑ Superellipse

- Implicit form

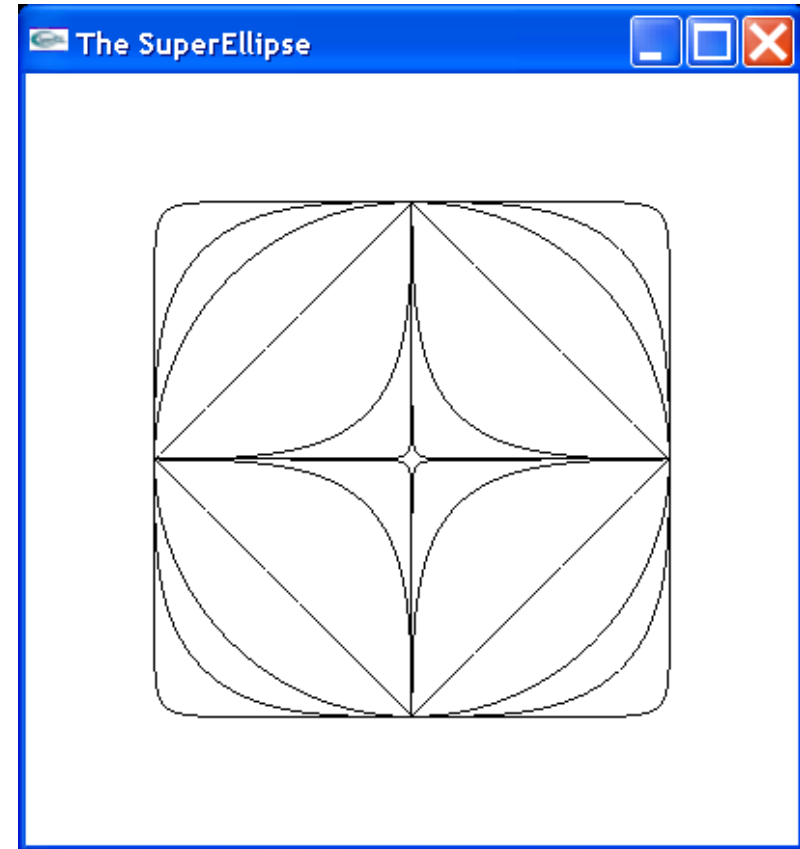
$$\left(\frac{x}{W}\right)^n + \left(\frac{y}{H}\right)^n = 1$$

- Parametric form

$$x(t) = W \cos(t) \left| \cos^{2/n-1}(t) \right|$$

$$y(t) = H \sin(t) \left| \sin^{2/n-1}(t) \right|$$

- $n = 2m/(2n+1)$
- $n < 1$ inward
- $n > 1$ outward
- $n = 1$ square



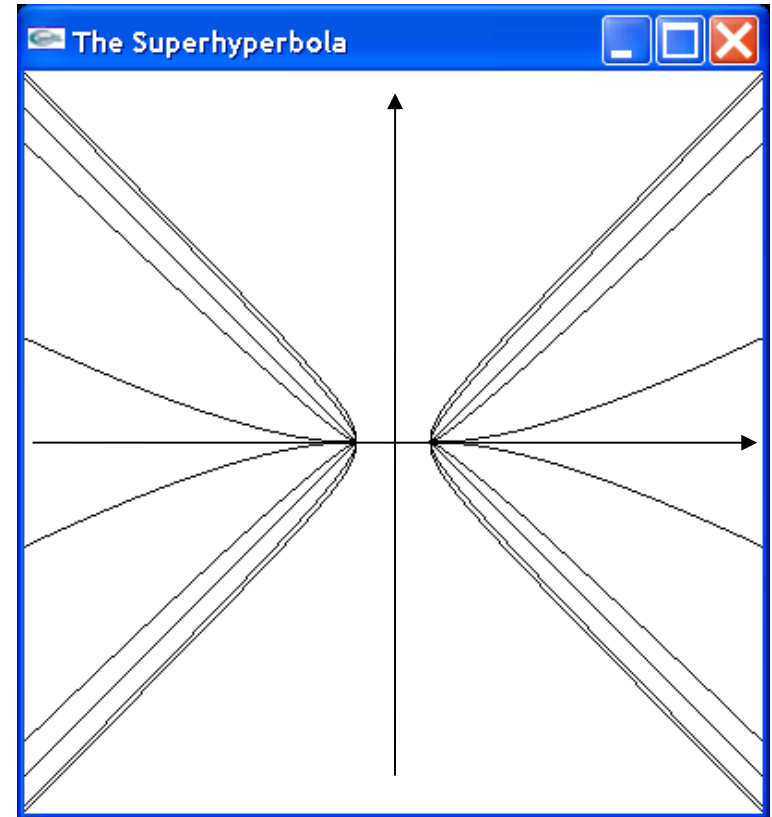
Parametric Form

❑ Superhyperbola

- Parametric form

$$x(t) = W \sec(t) \left| \sec^{2/n-1}(t) \right|$$
$$y(t) = H \tan(t) \left| \tan^{2/n-1}(t) \right|$$

- $n = 2m/(2n+1)$
- $n < 1$ inward
- $n > 1$ outward
- $n = 1$ line



Parametric Form

3D curves

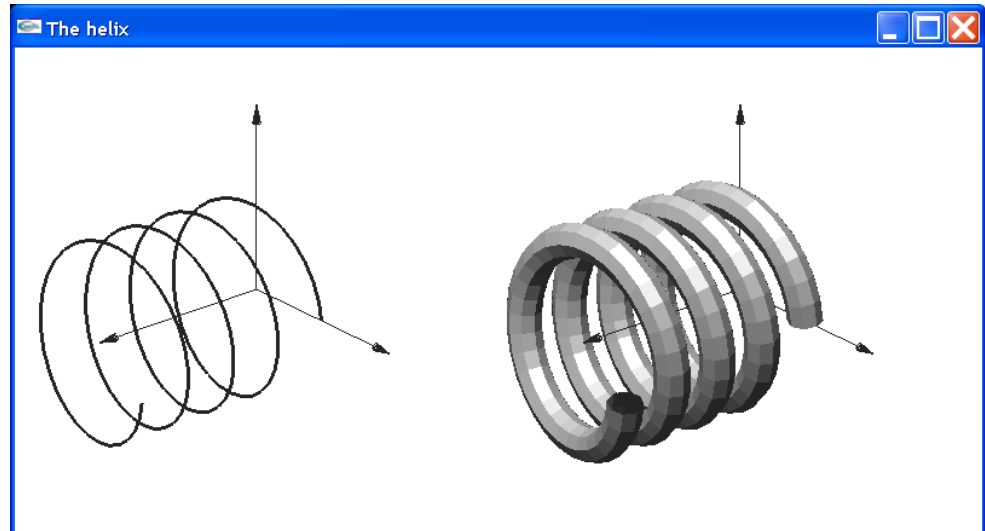
$$P(t) = (x(t), y(t), z(t))$$

Helix

$$x(t) = \cos(t)$$

$$y(t) = \sin(t)$$

$$z(t) = bt$$

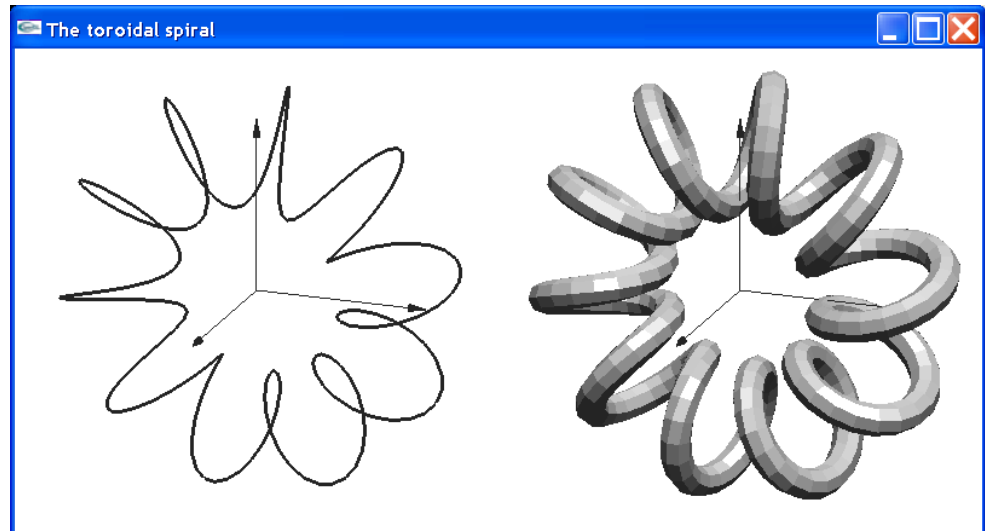


Toroidal spiral

$$x(t) = (a \sin(ct) + b) \cos(t),$$

$$y(t) = (a \sin(ct) + b) \sin(t),$$

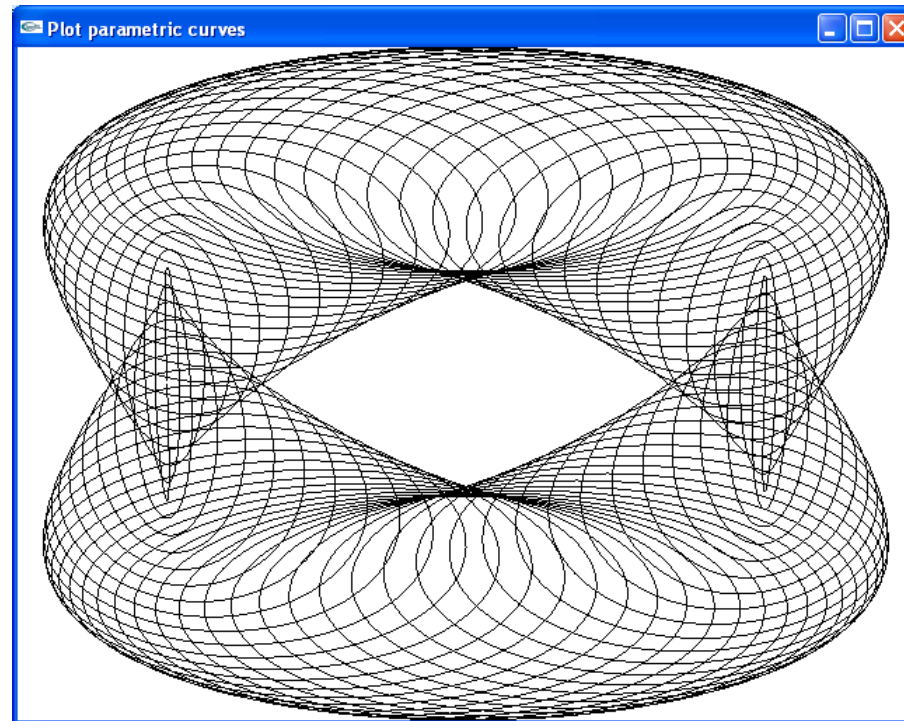
$$z(t) = a \cos(ct)$$



Parametric Form

$$x = \cos(t) - \cos(80 \cdot t) \cdot \sin(t);$$

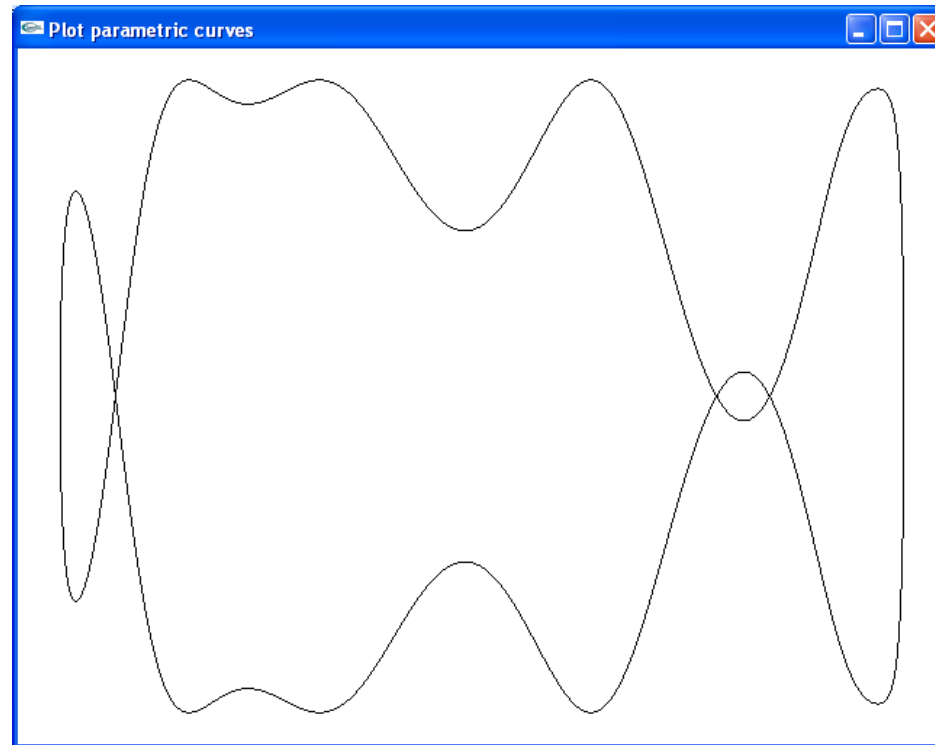
$$y = 2.0 \cdot \sin(t) - \sin(80 \cdot t);$$



Parametric Form

$$x = \cos(t);$$

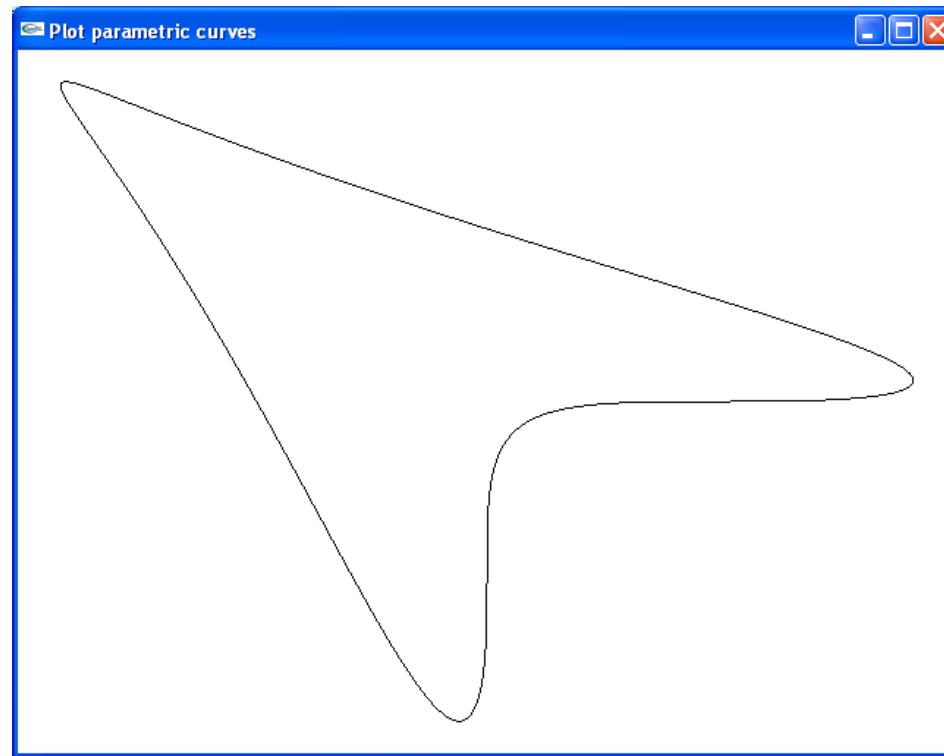
$$y = \sin(t + \sin(5.0*t));$$



Parametric Form

$$x = \sin(t + \sin(t));$$

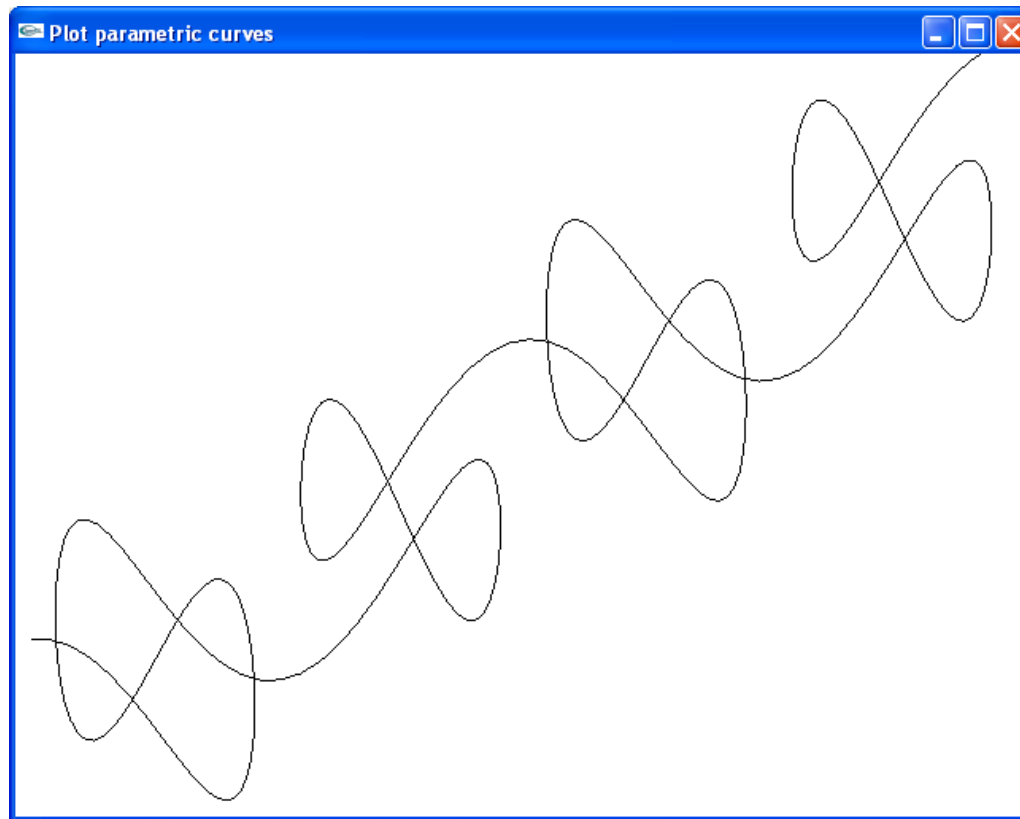
$$y = \cos(t + \cos(t));$$



Parametric Form

$$x = t + 2.0 \cdot \sin(2.0 \cdot t);$$

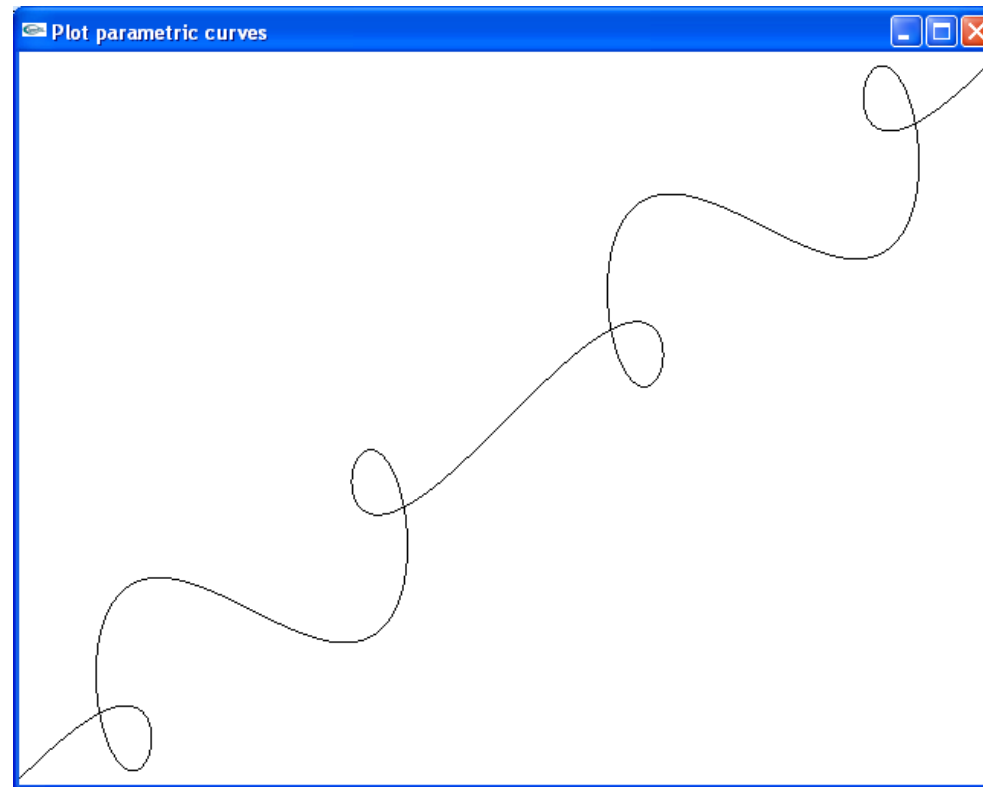
$$y = t + 2.0 \cdot \cos(5.0 \cdot t);$$



Parametric Form

$$x = t + \sin(2.0*t);$$

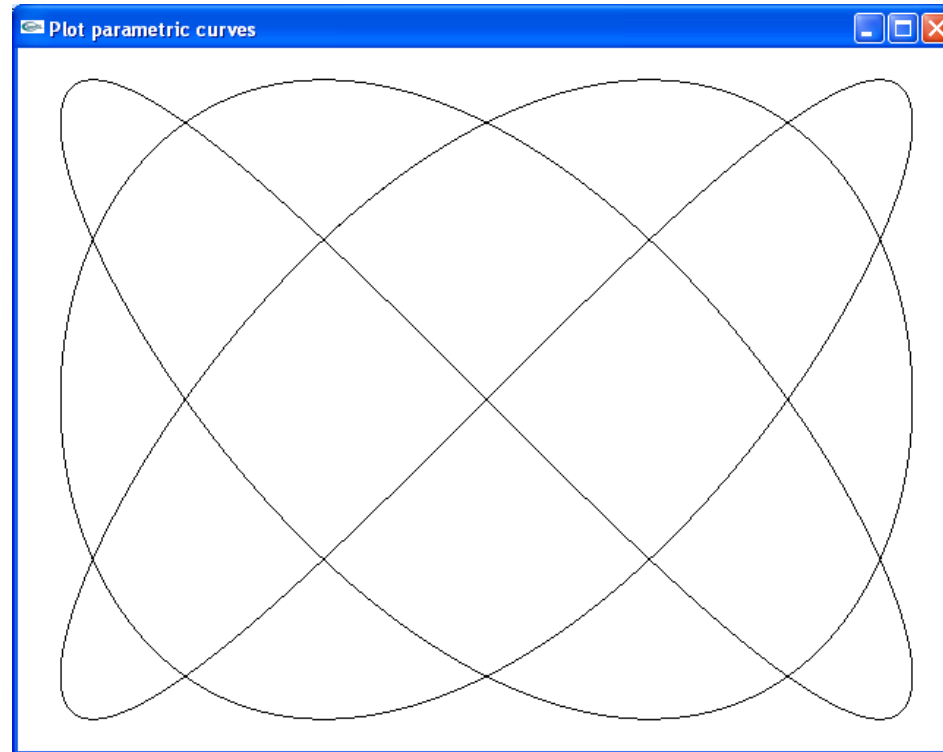
$$y = t + \sin(3.0*t);$$



Parametric Form

$$x = \sin(3.0 \cdot t);$$

$$y = \sin(4.0 \cdot t);$$

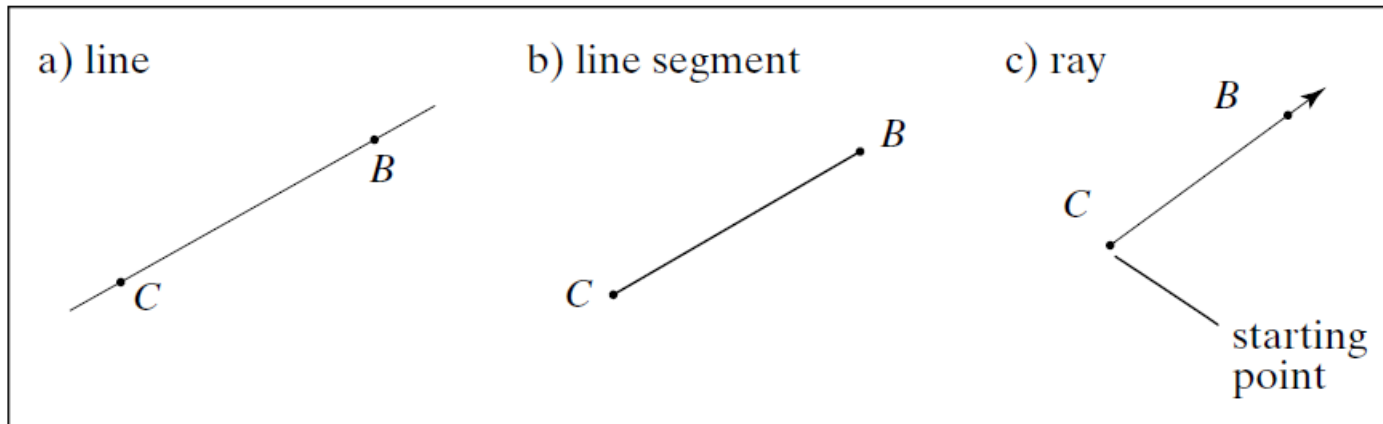


Line

$$y = 5x + 3 \quad (1)$$

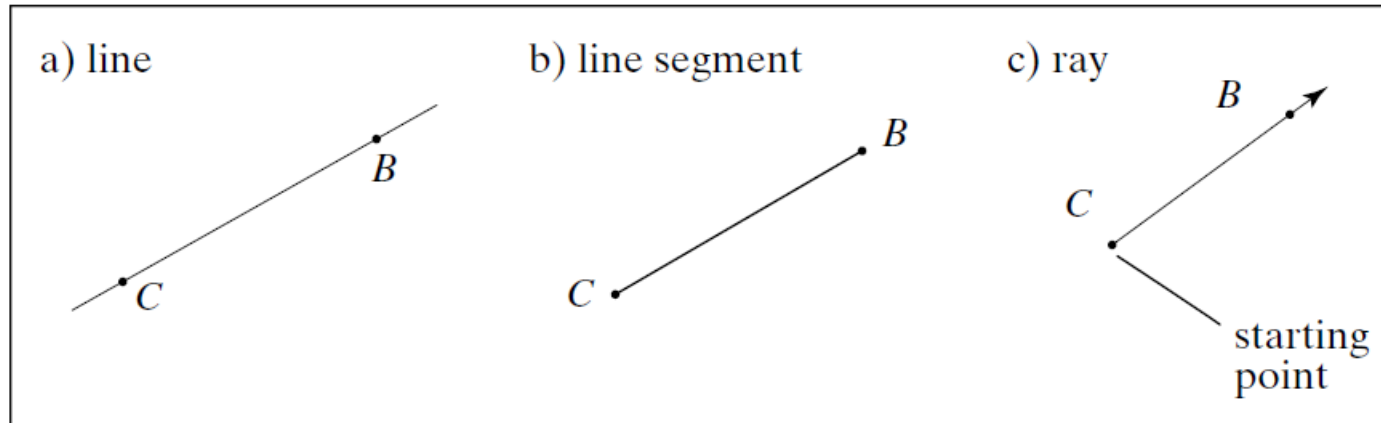
$$x = 1 \quad (2)$$

$$\frac{x - 3}{1} = \frac{y + 2}{6} = \frac{z - 3}{-2} \quad (3)$$



Line

□ Line, line segment, ray



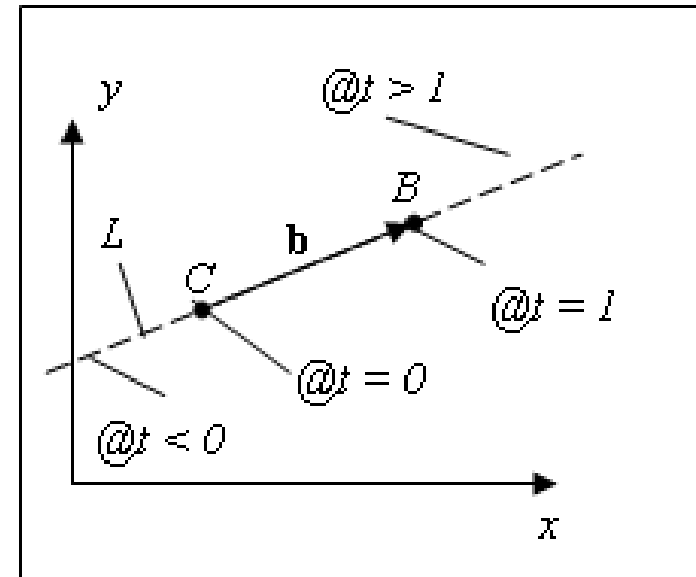
□ Parametric form

$$L(t) = C + \mathbf{b}t$$

Line segment, $0 \leq t \leq 1$

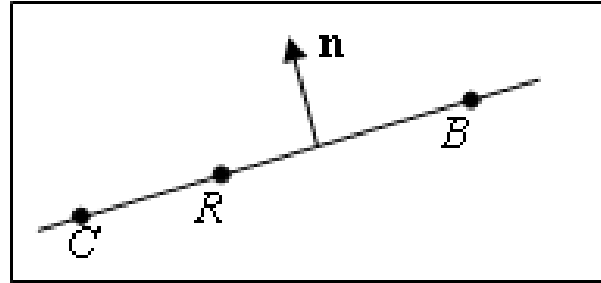
Ray, $0 \leq t < \infty$

Line, $-\infty \leq t < \infty$

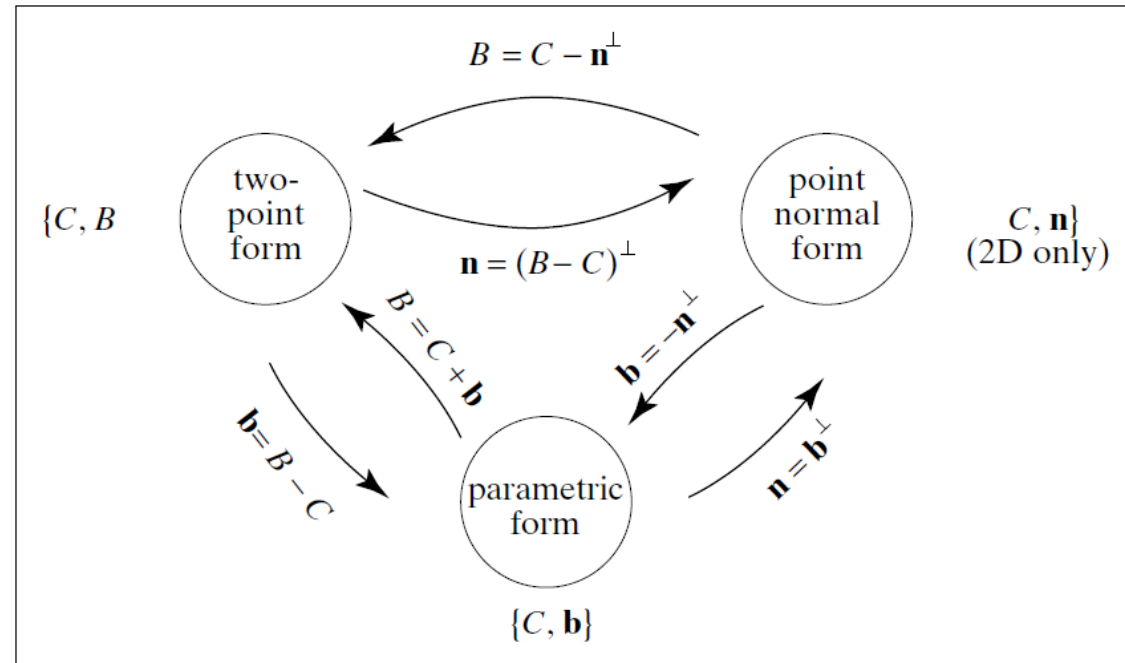


Line

□ Point-normal form: $\mathbf{n} \bullet (\mathbf{R} - \mathbf{C}) = 0$



□ Conversion



Plane

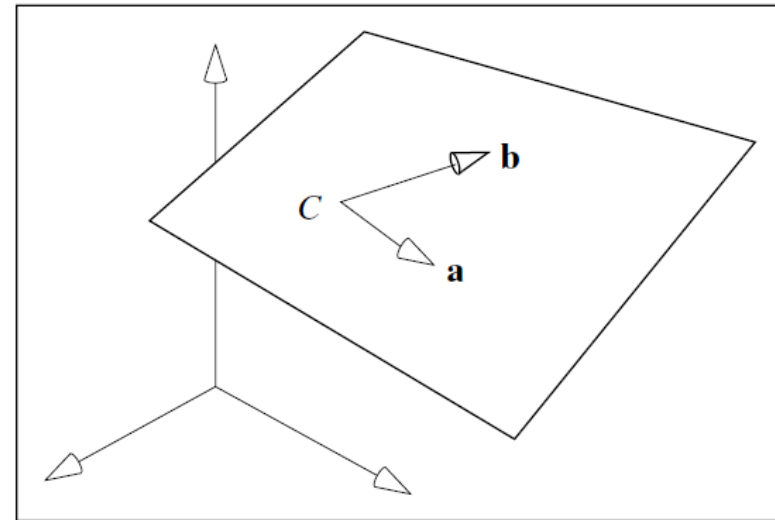
□ Parametric form:

$$P(s, t) = C + sa + tb$$

□ Point-normal form:

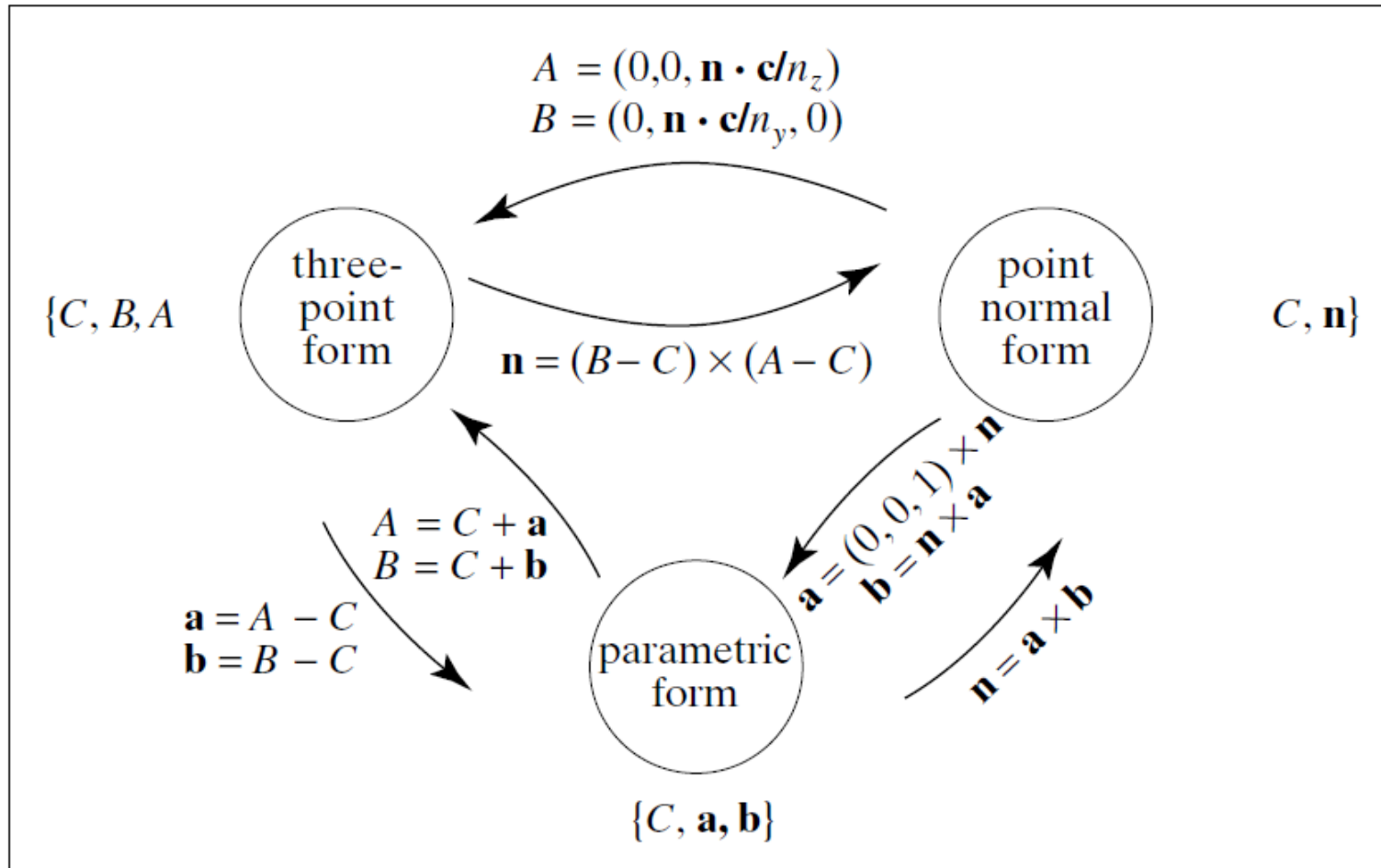
$$\mathbf{n} \bullet (\mathbf{R} - \mathbf{C}) = 0$$

$$\mathbf{n} = \mathbf{a} \times \mathbf{b}$$



Plane

□ Conversion



Some Examples

Intersection of two line segment

$$AB(t) = A + \mathbf{b}t ; CD(u) = C + \mathbf{d}u$$

Find t and u such as $A + \mathbf{b}t = C + \mathbf{d}u$

$$\mathbf{b}t = \mathbf{c} + \mathbf{d}u \text{ với } \mathbf{c} = C - A$$

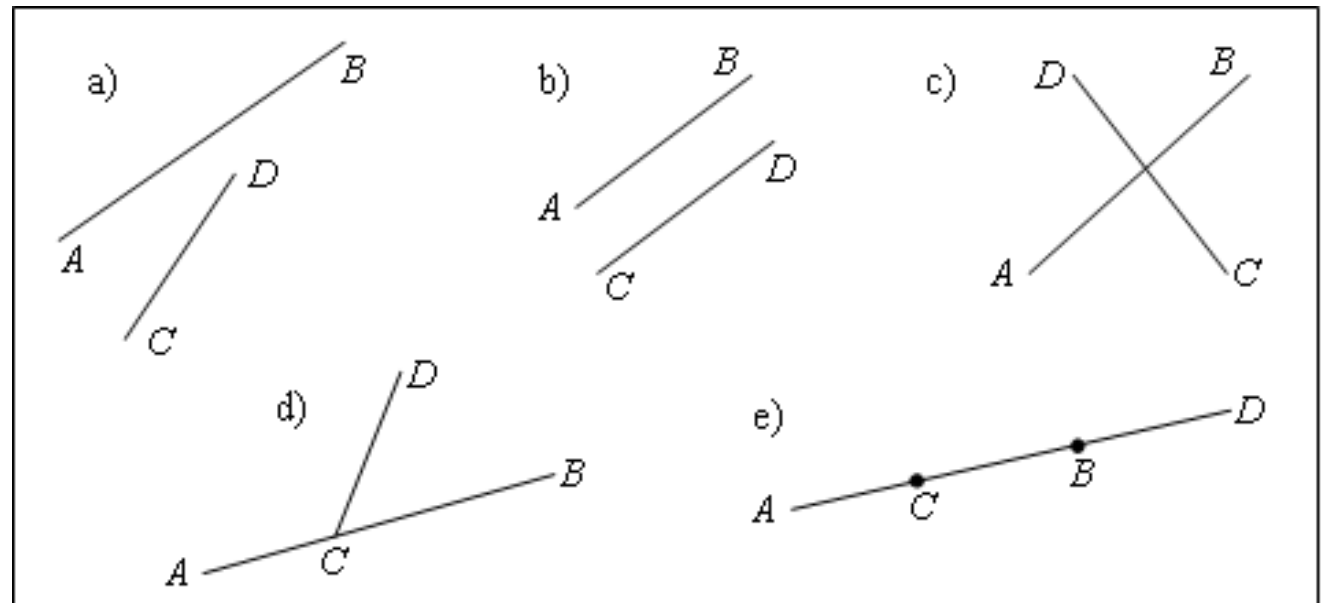
$$\mathbf{d}^\perp \bullet \mathbf{b}t = \mathbf{d}^\perp \bullet \mathbf{c}$$

$$\checkmark \mathbf{d}^\perp \bullet \mathbf{b} \neq 0.$$

$$t = \frac{\mathbf{d}^\perp \bullet \mathbf{c}}{\mathbf{d}^\perp \bullet \mathbf{b}}$$

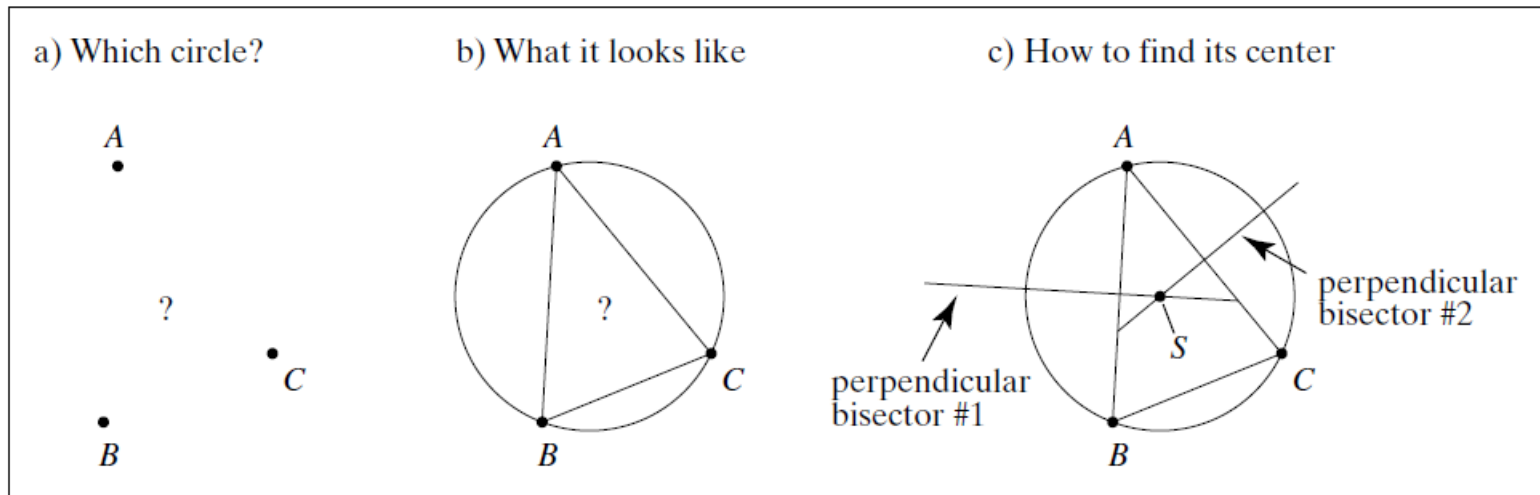
$$u = \frac{\mathbf{b}^\perp \bullet \mathbf{c}}{\mathbf{d}^\perp \bullet \mathbf{b}}$$

$$\checkmark \mathbf{d}^\perp \bullet \mathbf{b} = 0$$



Some Examples

□ The Circle through 3 points



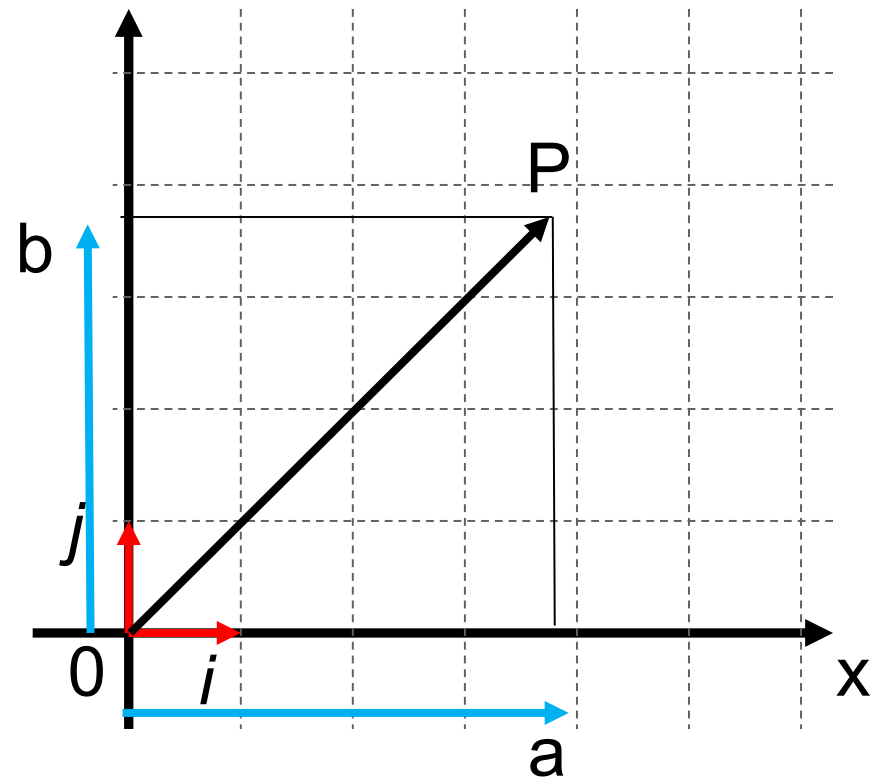
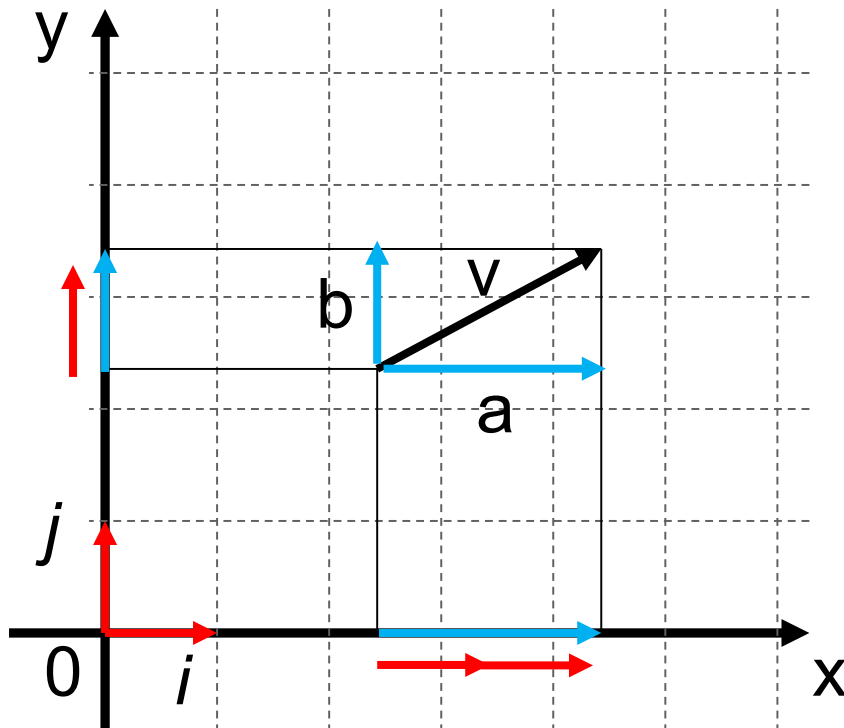
□ Perpendicular bisector $L(t) = \frac{1}{2}(A + B) + (B - A)^\perp t$

□ $\mathbf{a} = B - A$; $\mathbf{b} = C - B$; $\mathbf{c} = A - C$;

□ Perp. bisector AB: $A + \mathbf{a}/2 + \mathbf{a}^\perp t$; AC: $A - \mathbf{c}/2 + \mathbf{c}^\perp u$

□ $\mathbf{a}^\perp t = \mathbf{b}/2 + \mathbf{c}^\perp u \rightarrow t = \frac{1}{2} \frac{\mathbf{b} \bullet \mathbf{c}}{\mathbf{a}^\perp \bullet \mathbf{c}} \quad S = A + \frac{1}{2} \left(\mathbf{a} + \frac{\mathbf{b} \bullet \mathbf{c}}{\mathbf{a}^\perp \bullet \mathbf{c}} \mathbf{a}^\perp \right)$

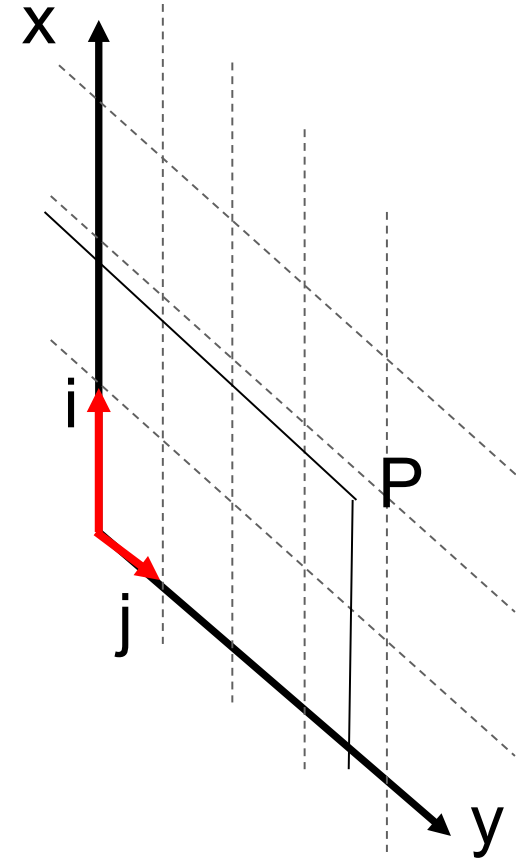
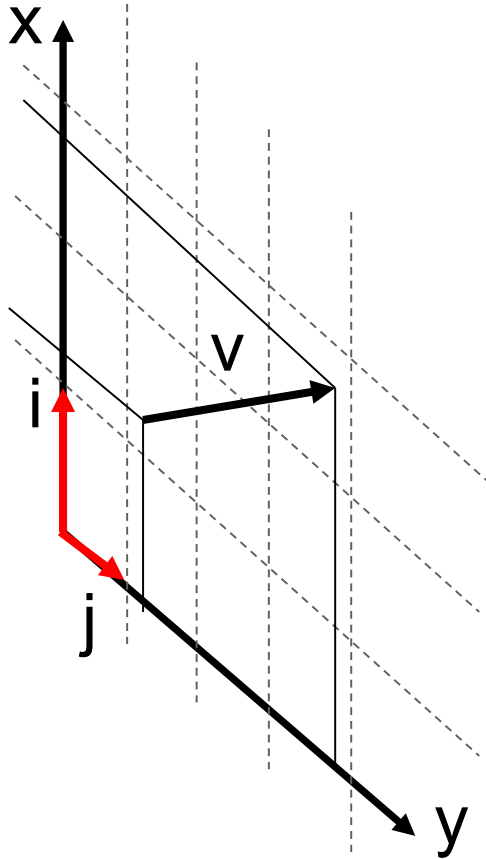
Representation



$$v = a + b = 2.1i + 1.2j \rightarrow v = (2.1, 1.2)$$

$$P = O + OP = O + a + b = O + 3.9i + 3.7j \rightarrow P = (3.9, 3.7)$$

Representation



$$v = 1.6i + 2.7j \rightarrow v = (1.6, 2.7)$$

$$P = (1.9, 3.6)$$

Representation

□ Frames

- Frame determined by (P_0, v_1, v_2, v_3)
- Within this frame, every vector can be written as

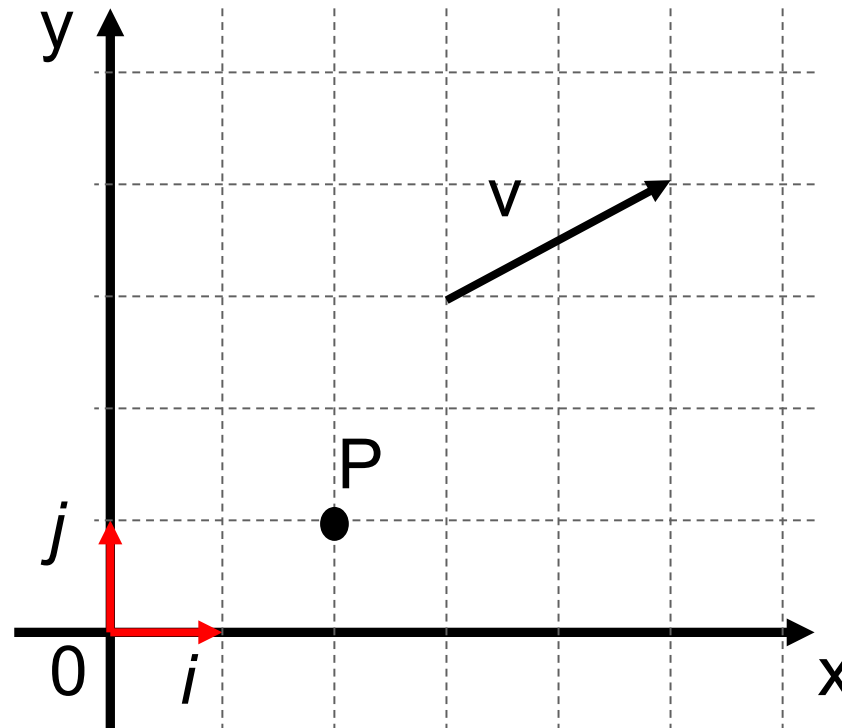
$$v = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$$

- Every point can be written as

$$P = P_0 + \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n$$

Representation

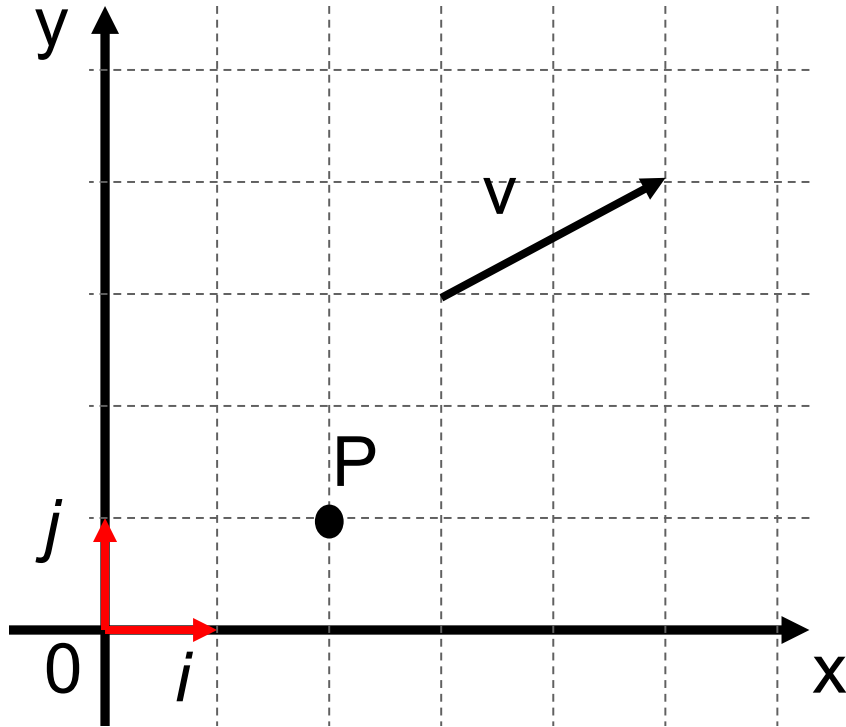
❑ Confusing Points and Vector



$$v = (2, 1); P = (2, 1)$$

Representation

❑ Confusing Points and Vector



$$\begin{aligned} v &= 2*i + 1*j = \\ &= 2*i + 1*j + 0*O \\ &= (2, 1, 0) \end{aligned}$$

$$\begin{aligned} P &= 2*i + 1*j + O \\ &= 2*i + 1*j + 1*O \\ &= (2, 1, 1) \end{aligned}$$

Representation

□ A Single Representation

If we define $0 \bullet P = \mathbf{0}$ and $1 \bullet P = P$ then we can write

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 = [\alpha_1 \ \alpha_2 \ \alpha_3 \ 0] [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ P_0]^T$$

$$\mathbf{P} = P_0 + \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 + \beta_3 \mathbf{v}_3 = [\beta_1 \ \beta_2 \ \beta_3 \ 1] [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ P_0]^T$$

Thus we obtain the four-dimensional *homogeneous coordinate* representation

$$\mathbf{v} = [\alpha_1 \ \alpha_2 \ \alpha_3 \ 0]^T$$

$$\mathbf{p} = [\beta_1 \ \beta_2 \ \beta_3 \ 1]^T$$

Representation

□ Homogeneous Coordinates

The homogeneous coordinates form for a three dimensional point $[x \ y \ z]$ is given as

$$\mathbf{p} = [x' \ y' \ z' \ w]^T = [wx \ wy \ wz \ w]^T$$

We return to a three dimensional point (for $w \neq 0$) by

$$x \leftarrow x'/w$$

$$y \leftarrow y'/w$$

$$z \leftarrow z'/w$$

If $w=0$, the representation is that of a vector

Note that homogeneous coordinates replaces points in three dimensions by lines through the origin in four dimensions

For $w=1$, the representation of a point is $[x \ y \ z \ 1]$

Representation

□ Change of Coordinate Systems

- Consider two representations of a the same vector with respect to two different bases. The representations are

$$\mathbf{a} = [\alpha_1 \ \alpha_2 \ \alpha_3]$$

$$\mathbf{b} = [\beta_1 \ \beta_2 \ \beta_3]$$

wher

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 = [\alpha_1 \ \alpha_2 \ \alpha_3] [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]^T$$

$$= \beta_1 \mathbf{u}_1 + \beta_2 \mathbf{u}_2 + \beta_3 \mathbf{u}_3 = [\beta_1 \ \beta_2 \ \beta_3] [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]^T$$

Representation

□ Change of Coordinate Systems

- Each of the basis vectors, u_1, u_2, u_3 , are vectors that can be represented in terms of the first basis

$$u_1 =$$

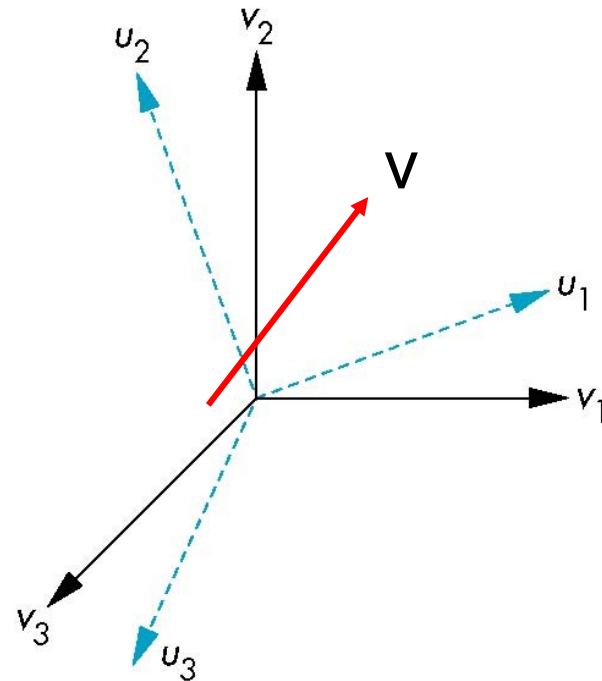
$$\gamma_{11}v_1 + \gamma_{12}v_2 + \gamma_{13}v_3$$

$$u_2 =$$

$$\gamma_{21}v_1 + \gamma_{22}v_2 + \gamma_{23}v_3$$

$$u_3 =$$

$$\gamma_{31}v_1 + \gamma_{32}v_2 + \gamma_{33}v_3$$



Representation

□ Change of Coordinate Systems

The coefficients define a 3 x 3 matrix

$$\mathbf{M} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} \end{bmatrix}$$

and the bases can be related by

$$\mathbf{a} = \mathbf{M}^T \mathbf{b}$$

Representation

Change of Coordinate Systems

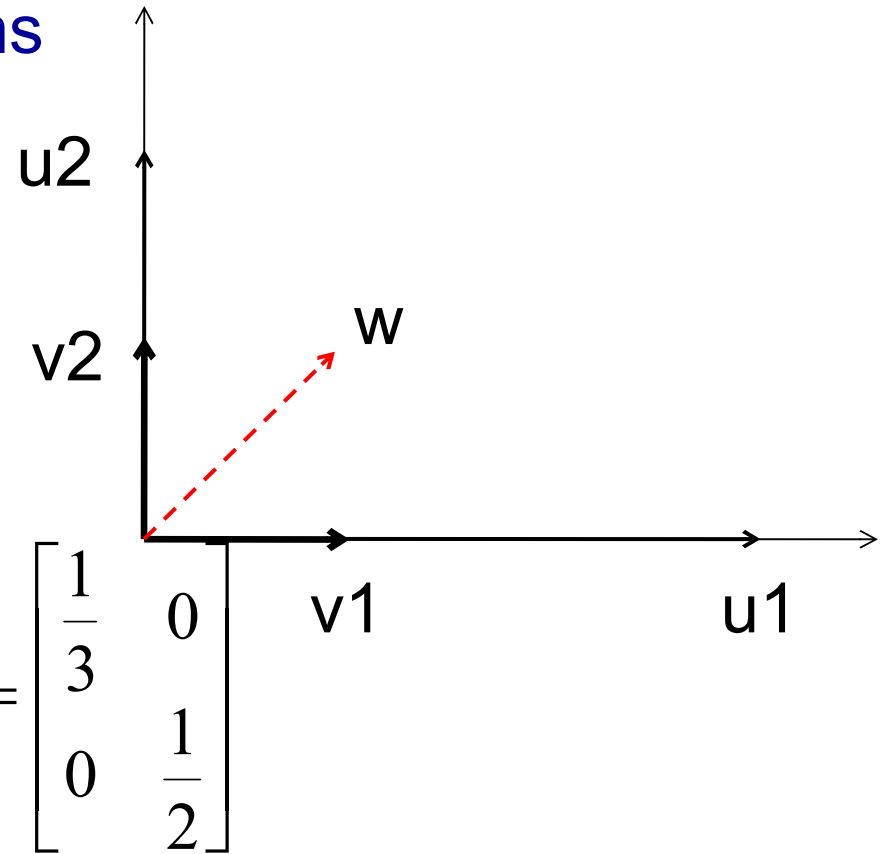
$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$u_1 = 3v_1$$

$$u_2 = 2v_2$$

$$M = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}; M^T = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}; (M^T)^{-1} = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; b = Ta = \begin{bmatrix} \frac{1}{3} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{2} \end{bmatrix}$$



Representation

□ Change of Coordinate Systems

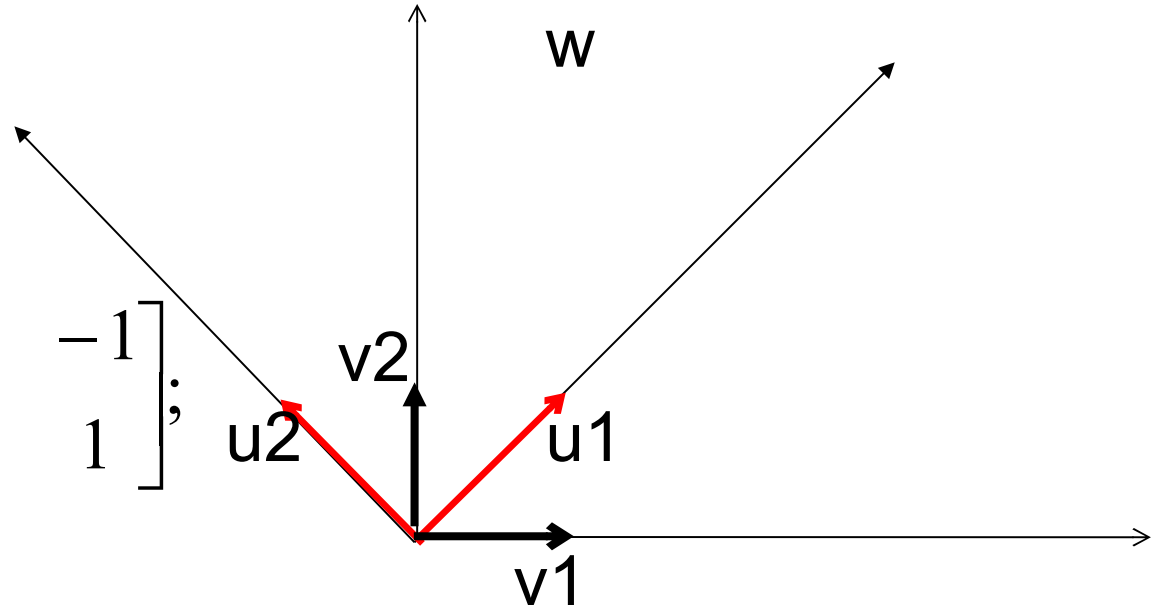
$$u_1 = v_1 + v_2$$

$$u_2 = -v_1 + v_2$$

$$M = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}; M^T = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix};$$

$$(M^T)^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$b = (M^T)^{-1} a = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



Representation

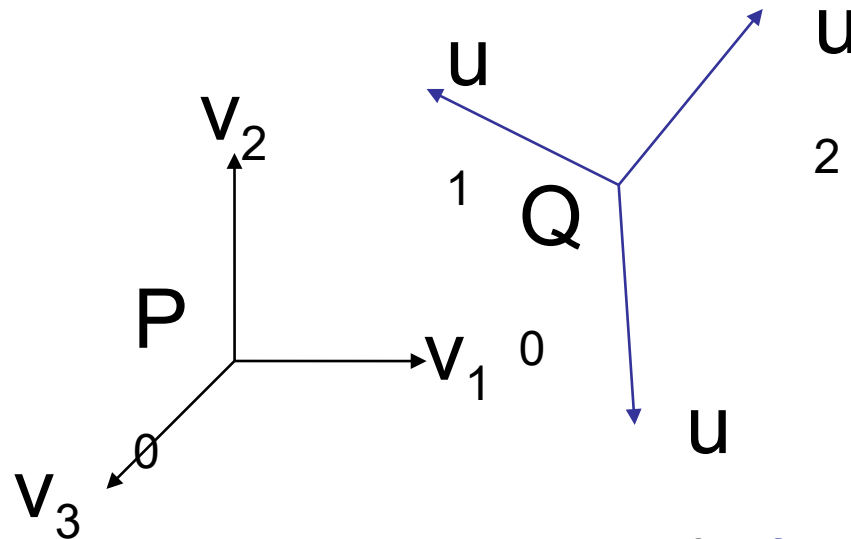
□ Change of Frames

- We can apply a similar process in homogeneous coordinates to the representations of both points and vectors

Consider two frames:

(P_0, v_1, v_2, v_3)

(Q_0, u_1, u_2, u_3)



- Any point or vector can be represented in either frame
- We can represent Q_0, u_1, u_2, u_3 in terms of P_0, v_1, v_2, v_3

Representation

□ Change of Frames

Extending what we did with change of bases

$$u_1 = \gamma_{11}v_1 + \gamma_{12}v_2 + \gamma_{13}v_3$$

$$u_2 = \gamma_{21}v_1 + \gamma_{22}v_2 + \gamma_{23}v_3$$

$$u_3 = \gamma_{31}v_1 + \gamma_{32}v_2 + \gamma_{33}v_3$$

$$Q_0 = \gamma_{41}v_1 + \gamma_{42}v_2 + \gamma_{43}v_3 + \gamma_{44}P_0$$

defining a 4 x 4 matrix

$$\mathbf{M} = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & 0 \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & 0 \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & 0 \\ \gamma_{41} & \gamma_{42} & \gamma_{43} & 1 \end{bmatrix}$$

Representation

□ Change of Frames

Within the two frames any point or vector has a representation of the same form

$\mathbf{a} = [\alpha_1 \ \alpha_2 \ \alpha_3 \ \alpha_4]$ in the first frame

$\mathbf{b} = [\beta_1 \ \beta_2 \ \beta_3 \ \beta_4]$ in the second frame

where $\alpha_4 = \beta_4 = 1$ for points and $\alpha_4 = \beta_4 = 0$ for vectors
and

$$\mathbf{a} = \mathbf{M}^T \mathbf{b}$$

The matrix $\mathbf{M}$ is 4 x 4 and specifies an affine transformation in homogeneous coordinates